ArtHOI: Taming Foundation Models for Monocular 4D Reconstruction of Hand-Articulated-Object Interactions

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https://arxiv.org/abs/2603.25791

Research Questions

- How can we reconstruct 4D hand-articulated-object interactions from a single monocular RGB video without requiring pre-scanned object templates or multi-view initialization?
- How can we reconcile inconsistencies between predictions from multiple foundation models (image-to-3D, depth estimation, pose estimation) to achieve physically plausible reconstructions?
- How can multimodal large language models be leveraged to reason about hand-object contact states for constraining optimization?

Core Idea

The paper tackles reconstructing 4D interactions between hands and articulated objects (like scissors, laptops) from single monocular videos - a highly ill-posed problem that existing methods can't handle without pre-scanned 3D models or multiple camera views. The key insight is to leverage multiple foundation models (image-to-3D, depth estimation, pose estimation, MLLMs) but carefully coordinate their outputs since naive combination fails due to inconsistent scales, coordinate systems, and physically implausible predictions. The method uses adaptive sampling to find the correct metric scale and pose for objects, and employs MLLMs to reason about contact states which then constrain optimization for physically plausible hand-object alignment.

Key Formula

Tracking Loss

$$L_{track} = \sum_{j \in S} \sum_{q \in W^k_{i,j}} \| z^k_{j,q} - (T^{pk}_i)^{-1} T^{pk}_j z^k_{i,q} \|$$

z^k_{j,q} are 3D tracks for part k, T^{pk}_i are per-part SE(3) transformations, W^k_{i,j} are visible tracks in both frames i and j, S is reference frame set. Enforces consistency of 3D tracking priors under visibility constraints for part-wise motion reconstruction.

Contact Loss

$$L_{contact} = \sum_{i \in C} \sum_{v_t \in T_i} \min_{v_o \in G^o_i} \| v_o - v_t \|^2$$

C is set of contact frames, T_i are MANO fingertip vertices for contacting fingers Y_i, G^o_i is object mesh at frame i. Minimizes distance from fingertips to closest object surface points, guided by MLLM contact reasoning for physically plausible hand-object alignment.

Key Figure

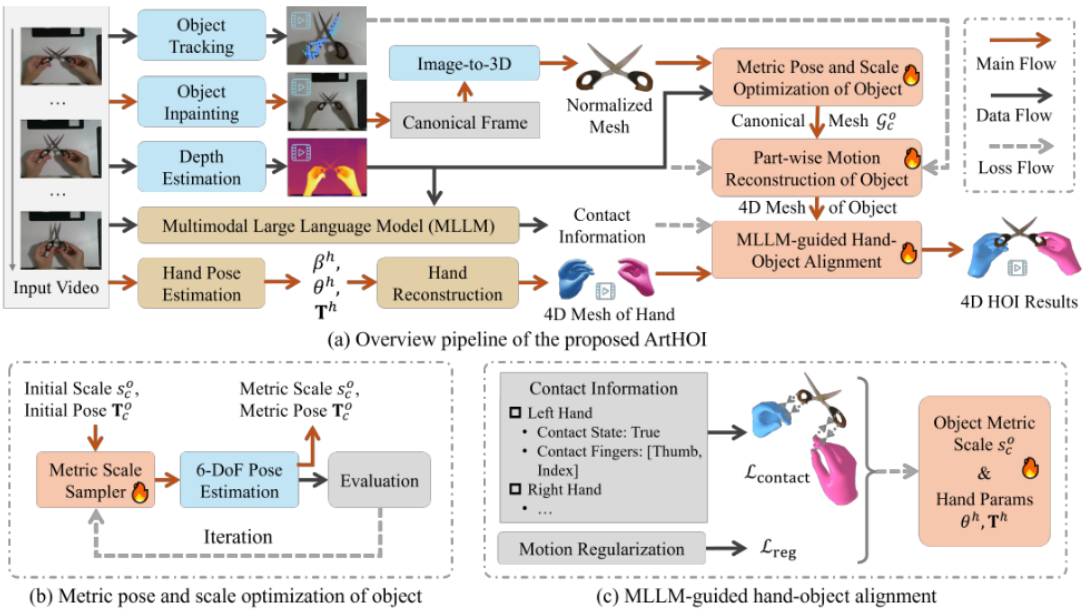

Figure 2. Pipeline of our ArtHOI. ArtHOI is an optimization-based framework (see subfigure (a)) that integrates and refines priors from multiple foundation models for monocular 4D reconstruction of human-articulated-object interactions. In particular, the proposed object's metric scale and pose optimization (see subfigure (b)) recovers 3D mesh in world space from a normalized one, while MLLM-guided hand-object alignment method (see subfigure (c)) promotes physically plausible hand-object mesh composition.

Figure 2: Shows the complete ArtHOI pipeline including the novel ASR method for metric scale/pose optimization and MLLM-guided

hand-object alignment, which are the key technical contributions.

Limitations

- Relies on accuracy of multiple foundation models - failures in depth estimation, segmentation, or image-to-3D generation can cascade through the pipeline
- Computational cost is high (~1 hour for 100 frames) due to iterative optimization stages, particularly part-wise motion reconstruction
- MLLM contact reasoning may still produce false positives/negatives, especially for ambiguous contact scenarios
- Limited to articulated objects that can be reasonably reconstructed by image-to-3D models, may fail on highly complex or transparent objects